SUITS 2026 Updates

Release 2/18:

TSS:

- Note that telemetry values in TSS will be adjusted after further testing, but general TSS functionality should not change much
- Below the EVA telemetry ranges in the eva-telemetry-ranges document, you can find details on how to handle error scenarios
- Suit resources currently start at nominal values but will be simulated to fill up during Egress. Similarly, ingress procedures will also be implemented in TSS soon.
- Some errors may only throw if all LTV errors are switched off. One of such errors should throw within the first 10 seconds of all LTV errors being switched off.
    - EVA control block will not indicate an error if you have switched DCU properly to resolve error even though simulation value may still be out of range

LTV Task Board

- LTV Task Board errors are subject to change
    - documents/procedures/LTV-repair-procedures document has information about some sample LTV repair procedures

DUST:

- LTV spawning distance has been decreased to a value we believe to be reasonable
    - LTV spawning location will not be constant in further updates, *so don't depend on the current location for searching!*
- TSS has to be running for DUST to work so once you start that up you type the IP address into the text box (no port, just xxx.xx.xxx.xx) and connect. To drive around hit Control+G and then use the arrows to drive around. Control+R to reset to origin position.
- There's also debug settings in the new settings menu (You can open with Escape or the P key) that lets you toggle indicators for the last known good position or rover position.

Maps:

- The maps in the documents->maps folder are from last year and may be updated

2/27:

Edited the data.h file

- It was not supposed to say ex. "ltv.errors.comms.nav_system"; rather, it should have just said ex. "ltv.errors.nav_system"

- Sorry for the confusion

3/3:

TSS:

- Overhaul documentation on PR, EV, and LTV procedures to more accurately reflect Test Week, standardize wording, and provide more information regarding potential errors

3/5

- Added second PING button. PING button now only allows you to ping once every 20 seconds. PING (unlimited) can be used for testing before coming to JSC and will work every second but will not be available at JSC. Additionally, PING (unlimited) feature also associated with command 2051.
- Backend restructuring of simulation (shouldn't affect students at all)
- A few bug fixes in error states and timing
- Added LTV_ERRORS.json file which can be retrieved with command number 3. Only the recovery mode code and instructions will be available until recovery mode is resolved. Once recovery mode is resolved, all instructions can be retrieved by the same command.
- Restructured UI so that ROVER team features on left side and SPACESUIT team features on right side
- Removed scrubber and fan from ROVER telemetry
- Added proper gradual temperature rise and fall for ROVER cabin heating and cooling
- Added document pr-team-ltv-search-info.pdf with information needed to calculate search radius *Note that values will be different for demonstration at JSC

3/11

- Updated LTV.json to latest version
- Initialized data in JSON files and switches at start of running program

4/16

- Note on the task board:
  - Before Recovery Mode error is resolved, requesting LTV_ERRORS.json file over UDP request will only send back Recovery Mode error code and procedures.
  - After Recovery Mode error is resolved, requesting LTV_ERRORS.json file over UDP request will send back Recovery Mode error code along with additional error codes
  - MAKE SURE YOU STILL CONTINUE TO CONSISTENTLY REQUEST INFORMATION FROM LTV_ERRORS.json until there are no more errors unresolved AS ADDITIONAL ERRORS MAY ARISE UPON RESOLVING OTHER ERRORS. I will add an example next week, so you can test your capabilities of checking for additional errors throughout the LTV error resolving process.

4/21

- Added photos of new DCU to README

4/24

- Example of LTV Error functionality implemented.

1. At first, sending the UDP command for LTV_ERRORS should just send you the recovery mode block:

```
{
    "error_procedures": [{
            "code": "4800",
            "description":  "Exit Recovery Mode (ERM)",
            "needs_resolved":   true,
            "procedures":   ["1. Investigate for prominent physical damage to any components before proceeding. Thi
        }, {
```

2. Upon successfully resolving Recovery Mode, the rest of the error blocks will show up:

```json
{
    "error_procedures": [{
            "code": "4800",
            "description":  "Exit Recovery Mode (ERM)",
            "needs_resolved":   true,
            "procedures":   ["1. Investigate for prominent physical damage to any components before proceeding. Thi
        }, {
            "code": "4509",
            "description":  "NAV Restart & Manual Return to Home ",
            "needs_resolved":   true,
            "procedures":   ["1. Locate the NAV Control Panel by identifying a large white rectangular component wi
        }, {
            "code": "1969",
            "description":  "Backup Fuse Error",
            "needs_resolved":   true,
            "procedures":   ["1. Locate the fuse box by identifying a large gray rectangular housing 2. Lift the li
        }, {
            "code": "3452",
            "description":  "Poor Comms RSSI",
            "needs_resolved":   true,
            "procedures":   ["1. Locate the antenna communications control component by identifying a square gray b
        }, {
            "code": "4968",
            "description":  "Subsystem Power Bus Error",
            "needs_resolved":   false,
            "procedures":   ["1. Locate the power distribution and diagnostic (PDD) component by identifying an irr
        }, {
            "code": "2441",
            "description":  "Comms Reboot Required Error",
            "needs_resolved":   false,
            "procedures":   ["1. Locate the power distribution and diagnostic (PDD) component by identifying an irr
        }, {
            "code": "2235",
            "description":  "Dust Sensor Error",
            "needs_resolved":   true,
```

3. Once you have resolved Poor Comms RSSI, notice that the Subsystem Power Bus error “needs_resolved” becomes true. If Poor Comms RSSI turns back to not resolved, the Subsystem Power Bus error “needs_resolved” will be false.

The same happens if Comms Reboot Required Error is resolved: once you have resolved the Comms Reboot Required Error, the Subsystem Power bus error “needs_resolved” becomes true. If Comms Reboot Required Error turns back to not resolved, the Subsystem Power Bus error “needs_resolved” will be false.

You will need to check for all LTV Task Board errors until there are no more LTV Task Board errors left to resolve. Once all LTV Task Board errors are resolved, no more new LTV Task Board errors will need resolved.

```
{
    "error_procedures": [{
            "code": "4800",
            "description":  "Exit Recovery Mode (ERM)",
            "needs_resolved":   false,
            "procedures":   ["1. Investigate for prominent physical damage to any components before proceeding. Thi
        }, {
            "code": "4509",
            "description":  "NAV Restart & Manual Return to Home ",
            "needs_resolved":   true,
            "procedures":   ["1. Locate the NAV Control Panel by identifying a large white rectangular component wi
        }, {
            "code": "1969",
            "description":  "Backup Fuse Error",
            "needs_resolved":   true,
            "procedures":   ["1. Locate the fuse box by identifying a large gray rectangular housing 2. Lift the li
        }, {
            "code": "3452",
            "description":  "Poor Comms RSSI",
            "needs_resolved":   false,
            "procedures":   ["1. Locate the antenna communications control component by identifying a square gray b
        }, {
            "code": "4968",
            "description":  "Subsystem Power Bus Error",
            "needs_resolved":   true,
            "procedures":   ["1. Locate the power distribution and diagnostic (PDD) component by identifying an irr
        }, {
            "code": "2441",
            "description":  "Comms Reboot Required Error",
            "needs_resolved":   false,
            "procedures":   ["1. Locate the power distribution and diagnostic (PDD) component by identifying an irr
        }, {
            "code": "2235",
            "description":  "Dust Sensor Error",
            "needs_resolved":   true,
```


5/7

- Updated pressurized-rover-procedure-timeline to only include relevant errors for this year's challenge
- Slightly updated some procedures in the ltv-repair-procedures document

5/8

- New Task Board photo with darker lighting
- New DCU photos with plain black background

5/9

- Uploaded latest version of ltv-repair-procedures.pdf in documents/procedures folder
- Updated LTV_ERRORS.json file with latest procedures
- Fixed a couple telemetry anomalies involving restart values and initial values

- Made DCU photos in README fit better on screen

5/17

- Updated LTV Last Nominal Position on pr-team-ltv-search-info document